

Backend Engineering Roadmap

6-Month Strategic Plan for Mobile-to-Backend Transition

A structured guide for senior engineers transitioning from Android development to distributed systems and microservices architecture.

Phase 1: The Monolithic Foundation

Month 1

- **Language Mastery:** Transitioning Kotlin skills to server-side (Spring Boot or Ktor).
- **API Fundamentals:** RESTful principles, HTTP/HTTPS lifecycle, and JSON serialization.
- **Data Persistence:** PostgreSQL fundamentals, ACID properties, and schema migrations (Flyway).
- **Project:** Build a high-performance Monolithic CRUD API with JWT-based security.

Phase 2: Persistence & Performance

Month 2

- **Caching:** Implementing Redis for session management and query caching.
- **Advanced SQL:** Optimization, indexing, and complex joins.
- **NoSQL Exploration:** Understanding MongoDB and Cassandra for horizontal scale.
- **Documentation:** Automated documentation with OpenAPI (Swagger).

Phase 3: The Microservices Pivot

Month 3

- **System Design:** Domain-Driven Design (DDD) and identifying Bounded Contexts.
- **Sync Communication:** Inter-service calls using Feign and gRPC.
- **Async Patterns:** Event-driven architecture with RabbitMQ or Apache Kafka.
- **Project:** Decompose the Phase 1 Monolith into orchestrated services.

Phase 4: Infrastructure & Dev-Ops

Month 4

- **Containerization:** Mastering Docker, multi-stage builds, and Docker Compose.
- **Service Discovery:** Dynamic routing with Netflix Eureka or Consul.
- **Gateway Pattern:** Implementing Spring Cloud Gateway for centralized ingress.

Phase 5: Observability & Resilience

Month 5

- **Fault Tolerance:** Circuit Breakers (Resilience4j) and Bulkheading.
- **The 3 Pillars:** Centralized logging (ELK), Metrics (Prometheus/Grafana), and Tracing (Jaeger).
- **Testing:** Integration testing and contract testing for distributed services.

Phase 6: Deployment & Scale

Month 6

- **Orchestration:** Kubernetes (K8s) basics—Pods, Deployments, and ConfigMaps.
- **CI/CD:** Building automated pipelines via GitHub Actions or GitLab CI.
- **Cloud Architecture:** Deploying to managed environments (AWS EKS or GCP).

Recommended Technology Stack

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Category	Tool/Framework
Language	Kotlin / Java 21
Framework	Spring Boot 3.x
Primary DB	PostgreSQL
Messaging	RabbitMQ or Kafka
Containers	Docker & Kubernetes
Observability	Grafana & Prometheus

Pro-Tip: Since you have experience with local hardware and Fedora, consider setting up a local K3s or Minikube cluster to simulate a production environment during Phase 4.